Shadows in tree geometries

Human Compression project

Abstract

For the coset geometry defined by the edge transpositions of a finite tree, the geometric intersection property holds in every residue precisely when the tree is a path. A common shadow in the positive case has a witness with at most two elements. We give the complete argument, including flag completion, residues and the transfer to inductively minimal geometries. The foundational classification of those geometries is an explicit retained input. These are established results; this text reorganizes the project's previously reviewed proof, rather than claiming a new classification.

1 Forest subgroups and their incidence geometry

Let T be a finite tree on a set Ω of $N \geq 2$ vertices, with edge set E. For $F \subseteq E$, let K_F be the subgroup of $\mathrm{Sym}(\Omega)$ generated by the endpoint transpositions of the edges in F. Set $H_e = K_{E\setminus\{e\}}$. The elements of type e in $\Gamma(T)$ are left cosets gH_e. Two such elements are incident when their cosets intersect; incidence is reflexive, and distinct elements of one type are not incident. A flag is a set of pairwise incident elements. The residue of a flag consists of its incident elements of the remaining types.

Lemma 1. *If C ranges over the components of (Ω, F), then*

$$K_F = \prod_C \mathrm{Sym}(C), \qquad K_F \cap K_D = K_{F\cap D}.$$

The coset system $\Gamma(T)$ is a thin, residually connected geometry on which $\mathrm{Sym}(\Omega)$ is flag-transitive. The residue at the standard flag of types $E \setminus F$ is the join of the geometries of the nontrivial components of (Ω, F).

Proof. Transpositions along a connected graph generate its full symmetric group: conjugating successive edge transpositions along a path produces the transposition of its endpoints. Distinct components have disjoint supports. A permutation in both K_F and K_D preserves every block of the common refinement of their component partitions. Two vertices are in such a block exactly when their unique tree path belongs to both forests, hence to $F \cap D$. This proves the two group formulas.

Root T, and let A_e be the component of $T - e$ away from the root. The coset gH_e identifies with the subset gA_e of Ω. For distinct types, incidence is precisely

$$|X_e \cap X_f| = |A_e \cap A_f|.$$

Indeed these intersection sizes and the two fixed subset sizes determine all four atoms of the pair; equal atom sizes give a permutation carrying one pair to the other. The A_e form a laminar family. Thus pairwise incidence forces exactly their nesting and disjointness relations on the X_e. Adjoin Ω above the selected sets and subtract each container's maximal selected proper subcontainers. The resulting atoms partition Ω and have the same sizes on the standard and target sides. Bijections of corresponding atoms combine to one permutation realizing the entire flag. Every flag therefore extends to a chamber, and flags of each fixed type form one orbit.

The stabilizer of the standard flag of types $E \setminus F$ is K_F. Its maximal parabolics for the remaining types are $K_{F\setminus\{e\}}$. Their direct product descriptions identify the residue with the claimed join; isolated vertices contribute trivial factors. The chamber stabilizer is $K_\varnothing = 1$, and each panel stabilizer has order two, so the geometry is thin. In any residue of rank at least two, the maximal parabolics collectively contain every remaining edge transposition, and hence generate the residue group. Their coset incidence graph is connected: products of subgroup elements give paths between standard and arbitrary cosets. This proves residual connectedness. □

For an element type i, write $\sigma_i(F)$ for the set of elements of type i incident with every member of F. The *intersection property* requires that for every element x and flag F, a nonempty $\sigma_i(x) \cap \sigma_i(F)$ be $\sigma_i(F')$ for a flag F' incident with x and F. This is required in the geometry and in every residue. It is distinct from the subgroup intersection identity of Lemma 1.

2 Explicit witnesses for Boolean shadows

Let $\mathcal{B}(\Omega)$ have the nonempty proper subsets of Ω as elements, cardinality as type, and comparability by inclusion as incidence.

Lemma 2. *Fix $0 < k < N$. For any family $\mathcal{U}$ of elements of $\mathcal{B}(\Omega)$, a nonempty common k-shadow has a witness flag of at most two elements, incident with every member of $\mathcal{U}$.*

Proof. If $\mathcal{U}$ contains a k-subset X, every common shadow element must equal X. Nonemptiness says that X is incident with the whole family, so $\{X\}$ is the required flag. Otherwise put

$$L = \bigcup_{\substack{X\in\mathcal{U}\\ |X|<k}} X, \qquad V = \bigcap_{\substack{X\in\mathcal{U}\\ |X|>k}} X,$$

with empty union $\varnothing$ and empty intersection Ω. The common shadow is exactly

$$\{Y \subseteq \Omega : |Y| = k,\ L \subseteq Y \subseteq V\}. \tag{1}$$

Nonemptiness gives $L \subseteq V$ and $|L| \le k \le |V|$. If either cardinality bound is equality, the shadow has a unique element, which again supplies a singleton flag incident with the whole family. In the remaining case use $\{L, V\}$, omitting $\varnothing$ and Ω. Its members are proper subsets of different sizes and form a flag. Every smaller query member is contained in L, and every larger one contains V, so both witness elements are comparable with every query member. Its shadow is precisely (1), including when one or both boundary sets were omitted. □

Lemma 3. *The geometry $\mathcal{B}(\Omega)$ satisfies the intersection property in all residues. Joins of geometries with that property also satisfy it.*

Proof. Apply Lemma 2 to $\mathcal{U} = \{x\} \cup F$ for the property in $\mathcal{B}(\Omega)$. A fixed chain, completed by the formal endpoints $\varnothing$ and Ω, divides its residue into the proper intervals between successive members. An interval (A, B) is the Boolean geometry on $B \setminus A$, via $X \mapsto X \setminus A$; the types are shifted by $|A|$. All elements in different intervals are incident, so the residue is their join. Empty intervals add no types. A rank-one factor has only empty, full or singleton shadows; the empty flag witnesses a full shadow, and its unique point witnesses a singleton.

For completeness, in a join fix a type i in one factor. Only query elements of that factor constrain its shadow. If x belongs to it, use the factor's intersection property with the part of F in that factor. If x belongs to another factor, the part of F in the target factor is itself a witness, including when it is empty. The witness is automatically incident with all query elements in other factors. Residues of a join are joins of factor residues, proving the assertion also at every residue. □

Theorem 4. *The geometry $\Gamma(T)$ satisfies the intersection property if and only if T is a path.*

Proof. For a path, root at one endpoint. Its cuts A_e form a chain with sizes $1, \dots, N-1$. The subset description in Lemma 1 identifies its incidence with comparability, so $\Gamma(T) = \mathcal{B}(\Omega)$. Lemma 3 proves the positive assertion.

A tree which is not a path has three edges incident at a common vertex. Keep precisely those edges in Lemma 1. The resulting residue is the geometry of the claw, with three types a, b, c, each a copy of $\{1, 2, 3, 4\}$. Elements of different types are incident exactly when their labels differ. Take $x = (b, 1)$ and $F = \{(b, 2)\}$, with target type a. Their common shadow is $\{(a, 3), (a, 4)\}$. A witness cannot contain an a-element, which would give at most a singleton shadow, or a b-element, which cannot be incident with both distinct b-elements. It can contain at most one c-element. The empty flag has four shadow elements, whereas any c-element has three. Neither has the required shadow. Thus this residue fails the property. □

3 Transfer to inductively minimal geometries

We retain the foundational classification of Buekenhout, Cara and Dehon, in the precise form recalled by Buekenhout and Cara [BC98]: an inductively minimal pair of positive rank r has group $\mathrm{Sym}(r+1)$; its connected diagram has no induced cycle longer than three, each edge belongs to a unique maximal clique, and each vertex belongs to one or two maximal cliques. Every such connected diagram determines one pair up to isomorphism. The one-vertex diagram is treated separately. The proof of this foundational classification is not replaced here.

These diagrams are precisely line graphs of trees. To see the required direction, form a vertex for each maximal clique. A diagram vertex in two cliques gives an edge between their clique vertices; a vertex in one clique gives an edge to a new leaf. Distinct cliques meet in at most one vertex, because a common edge would violate uniqueness of its clique. The constructed graph is simple and connected, and its line graph is the original diagram. A shortest cycle of length at least four would give an induced cycle on its edge labels in the diagram. A triangle would place three pairwise adjacent labels in one clique, contradicting unique ownership of the pairwise edges by the three cliques already represented at the cycle vertices. The constructed graph is therefore a tree. Its reconstruction is inverse to taking the nontrivial vertex stars of a tree as line-graph cliques. The one-vertex diagram corresponds to the one-edge tree.

For the geometry in Lemma 1, two remaining edges give either a six-cycle residue or $K_{2,2}$, according as they meet or are disjoint. Its diagram is consequently the line graph of T. Every residue with connected diagram and m remaining types comes from a single subtree on $m+1$ vertices, and its group is $\mathrm{Sym}(m+1)$. It is minimal under the source bound $|G| \le (m+1)!$. Thus $\Gamma(T)$ is inductively minimal. The retained uniqueness identifies every inductively minimal pair with one of these tree geometries.

Corollary 5. *An inductively minimal geometry has the intersection property exactly when its diagram is linear. In particular this holds for the* $\mathrm{Sym}(n)$ *geometries of rank* $n-1$ *with* $2 \le n \le 7$.

Proof. Apply the retained classification, the construction just proved, and Theorem 4. The line graph of a tree is a path exactly when the tree itself is a path: a vertex of degree at least three would give a triangle in the line graph. □

Attribution and verification scope

Section 5 of [JL04] reports the symmetric-group observation recovered by this corollary. That paper already attributes the all-degree statement to Cara and Leemans, whose published article is *On inductively minimal geometries that satisfy the intersection property*, J. Combin. Theory A

111 (2005), 327–330. The latter full text was unavailable in the fixed intake; it remains archived and receives no whole-original credit from this corollary.

The forest and subset arguments above were already written and independently reviewed in the project's earlier tree-cut derivation; the forest theorem and the foundational transfer also occur in the current Topic B papers. The present text makes those proof dependencies explicit within this application. It claims no new historical theorem or open-problem solution. The inherited Lean module `BooleanShadow.lean` proves the interval formula, the singleton case and the existence of a proper witness of size at most two for arbitrary finite query families. It does not formalize the tree geometry, its residues, the claw obstruction or the foundational classification. Its recorded axioms are `propext`, `Classical.choice` and `Quot.sound`. The earlier finite checks and compiler receipt are retained without a redundant execution; the complete argument here relies on the written proofs, not on extrapolation from finite cases.

References

[BC98] F. Buekenhout and P. Cara, Some properties of inductively minimal geometries, *Bull. Belg. Math. Soc. Simon Stevin* 5 (1998), 213–219. The foundational 1997 classification is retained as recalled there, not claimed independently rederived.

[CC02] P. J. Cameron and P. Cara, Independent generating sets and geometries for symmetric groups, *J. Algebra* 258 (2002), 641–650. The transposition-tree geometry is an established model.

[JL04] P. Jacobs and D. Leemans, An algorithmic analysis of the intersection property, *LMS J. Comput. Math.* 7 (2004), 284–299, Section 5. This complete source and its supplied archive are in the fixed accessible corpus.